M. BASHEER ALI

STORAGE QA ENGINEER
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PROFESSIONAL PROFILE

Storage QA Engineer with 3 years of experience validating enterprise storage arrays, SSD/HDD/NVMe drives, RAID, Fibre Channel, and iSCSI environments. Skilled in Python/Pytest automation, Linux debugging, FIO performance testing, and storage protocol validation. Experienced in developing automation frameworks that improve regression efficiency and ensure high-quality enterprise storage releases.

PROFESSIONAL SUMMARY

- Proficient in Python with strong knowledge of OOP concepts, data structures, file handling, exception handling, regular expressions, and scripting.
- Experienced in developing and maintaining Python/Pytest automation frameworks for storage validation, regression testing, log collection, and result analysis.
- Proficient in Python modules including os, sys, subprocess, argparse, hashlib, logging, datetime, json, re, and Paramiko for automation, remote execution, and log analysis.
- Hands-on experience in validating enterprise storage platforms, focusing on drive qualification, compatibility, firmware, reliability, performance, and fault recovery.
- Strong understanding of Functional, Integration, System, Regression, Performance, Stress, and Re-testing methodologies with knowledge of SDLC, STLC, Agile, and Defect Life Cycle.
- Experienced in designing, executing, and automating test cases for drive validation, RAID validation, firmware validation, and storage feature testing.
- Strong knowledge of enterprise storage technologies including HDD, SSD, NVMe, Storage Arrays, JBOD, DAS, SAN, and NAS.
- Familiar with storage protocols including SATA, SAS, SCSI, NVMe, iSCSI, and Fibre Channel (FC).
- Skilled in storage performance analysis using FIO, evaluating IOPS, throughput, latency, bandwidth, and queue-depth metrics.
- Experienced in storage diagnostics using smartctl, nvme-cli, storcli, lsblk, lspci, dmesg, and Linux system logs for troubleshooting and root-cause analysis.
- Good understanding of SSD technologies including wear leveling, TRIM, garbage collection, write amplification, over-provisioning, endurance, and SMART health monitoring.
- Strong knowledge of RAID technologies including RAID levels, Hot Spare, Online RAID Level Migration (ORLM), Capacity Expansion, Consistency Check, Rebuild, and Hot Swapping.
- Proficient with Linux, Shell Scripting, Git, Jenkins, JIRA, PyCharm, PuTTY, VMware, and x86 server firmware components including BIOS, BMC, and CPLD.
- Leveraged GitHub Copilot and Claude to accelerate automation development, log analysis, and debugging.

TECHNICAL SKILLS

Storage Management: DAS, SAN, Storage Arrays
Storage Technologies: HDD, SSD, SATA, SAS, iSCSI, RAID, NVMe
Storage & Testing Tools: IOmeter, FIO, mdadm, storcli, smartctl, IPMI, JIRA
AI & Automation Tools: Claude, GitHub Copilot, OpenAI
Programming & Automation: Python, Pytest, Shell Scripting, Jenkins, Git,
Testing: Manual & Automation Testing, SDLC, STLC, Agile Methodology, Bug Life Cycle, Bug Tracking
Servers & Virtualization: BIOS, BMC, CPLD, VM's, Snapshots, Cloning, HA, FT, DRS
Operating Systems: RHEL 9.4, Windows (2022, 2025), VMWare ESXI (8.0, 9.0)

PROFESSIONAL EXPERIENCE

Company Name – Xenops Technologies Private Limited Sep 2023 - present
Role: Storage QA Engineer

PROJECTS

Enterprise HDD/SSD/NVMe Drive Qualification – DDN A3I

Client: DDN (DataDirect Networks) Aug 2025 – present

Project Description

Performed enterprise drive validation for the DDN A3I storage platform, focusing on HDD, SSD, and NVMe qualification, compatibility, performance, reliability, firmware, and fault-recovery testing. Developed Python-based automation and leveraged AI-assisted tools for test development, log analysis, troubleshooting, and validation efficiency.

Roles & Responsibilities

- Validated enterprise storage platform components including controllers, BIOS, BMC, CPLD, SSD/HDD/NVMe, HBAs, PCIe, network interfaces, power supplies, and storage enclosures across hardware and firmware configurations.
- Validated BIOS/BMC functionality and integration, including boot, device enumeration, configuration persistence, system health sensors, event logs, remote console, power control, firmware upgrade/downgrade, reset, and recovery.
- Executed platform fault and recovery scenarios including controller/BMC/BIOS resets, reboots, power cycles, firmware failures, and device enumeration recovery.
- Performed RAID, storage pool, volume/LUN, hot-spare, degraded-mode, rebuild, recovery, firmware, compatibility, and fault-injection validation.
- Validated FC/iSCSI connectivity, multipathing, path failover, controller failover/failback, and continuous I/O during reboot and power-cycle scenarios.
- Developed and maintained Python/Pytest automation frameworks using reusable fixtures, parametrization, assertions, utilities, configuration management, and modular test components for scalable storage validation.
- Automated storage health checks, drive discovery, RAID validation, FIO execution, firmware verification, log collection, and result validation using Python and Pytest, reducing repetitive validation effort by 50%.
- Implemented Python automation using modules such as os, sys, subprocess, json, datetime, and Paramiko for remote execution, process management, log parsing, and test orchestration.

- Integrated automated test execution with Jenkins and Git, enabling scheduled regression runs, automated result collection, failure reporting, and repeatable validation across firmware releases.
- Executed 200+ regression and feature test cases across multiple firmware releases and validated 15+ FC/iSCSI configurations.
- Analyzed platform, controller, drive, OS, BMC, and performance logs to identify failures, error patterns, and root causes.
- Leveraged ChatGPT/OpenAI and GitHub Copilot for Python automation development, Pytest test generation, code optimization, troubleshooting, and AI-assisted log analysis.

Enterprise Storage Platform & Feature Validation – DDN IntelliFlash

Client: DDN (DataDirect Networks) **(**Sep 2023 – Aug 2025)

Project Description

Worked on the validation of the DDN IntelliFlash enterprise all-flash storage platform, ensuring hardware, firmware, and software quality. Performed platform and feature validation across multiple storage configurations and firmware versions. Executed functional, regression, compatibility, and performance testing to verify reliability and stability.

Roles & Responsibilities

- Validated enterprise storage components including controllers, BIOS, BMC, CPLD, SSD/HDD/NVMe, HBAs, PCIe, networking, power supplies, and storage enclosures across hardware and firmware configurations.
- Validated BIOS/BMC functionality including boot, device enumeration, sensors, event logs, remote console, power control, firmware updates, reset, and recovery.
- Executed platform fault and recovery scenarios covering controller/BMC/BIOS resets, reboots, power cycles, firmware failures, and device recovery.
- Performed RAID, storage pool, volume/LUN, hot-spare, degraded-mode, rebuild, recovery, compatibility, firmware, and fault-injection validation.
- Validated FC/iSCSI connectivity, multipathing, path failover, controller failover/failback, and continuous I/O during reboot and power-cycle scenarios.
- Conducted FIO-based performance, stress, and endurance testing, analyzing IOPS, throughput, latency, block sizes, queue depths, and workload patterns.
- Developed and maintained Python/Pytest automation frameworks with reusable fixtures, parametrization, assertions, utilities, and modular test components for scalable storage validation.
- Automated storage health checks, drive discovery, RAID validation, FIO execution, firmware verification, log collection, and result analysis, reducing manual validation effort by 50%.
- Implemented Python automation using os, sys, subprocess, json, datetime, hashlib, and Paramiko for remote execution, process management, log parsing, and test orchestration.
- Integrated automation with Git/Jenkins for scheduled regression execution and leveraged ChatGPT/OpenAI and GitHub Copilot for test generation, code optimization, debugging, and AI-assisted log analysis.

EDUCATION

Bachelor of Science (Computer Science) | PRGC(A)**.** **2020 – 2023**